

Carbon and its Compounds

Covalent bonding, allotropes of carbon, saturated and unsaturated hydrocarbons, homologous series, functional groups, and soaps and detergents, with examples and practice questions.

Introduction

Carbon is present in all living things. It is found in food, clothes, medicines and fuels. Carbon forms a very large number of compounds. This is possible because carbon can form strong bonds with many other elements. In this chapter we study how carbon bonds, what types of carbon compounds exist, and how they are named and used.

Covalent Bonding

Covalent bonding is the sharing of electrons between atoms. Carbon always forms covalent bonds. It does not form ionic bonds easily. There are two reasons for this.

First, carbon has four electrons in its outermost shell. It needs four more electrons to complete its octet. It cannot lose four electrons because that needs a lot of energy. It cannot gain four electrons either, because the nucleus with only six protons cannot hold ten electrons.

So carbon shares its four valence electrons with other atoms. Each shared pair forms a covalent bond.

Covalent Bond

A bond formed by sharing of electrons between two atoms so that both atoms get a stable noble gas configuration.

Examples:

- **Hydrogen molecule (H₂):** Each H atom shares one electron. One single bond is formed. H-H.
- **Chlorine molecule (Cl₂):** Each Cl shares one electron. One single bond.
- **Oxygen molecule (O₂):** Each O shares two electrons. A double bond is formed. O=O.
- **Nitrogen molecule (N₂):** Each N shares three electrons. A triple bond is formed. N≡N.
- **Methane (CH₄):** Carbon shares its four electrons with four hydrogen atoms. Four single bonds are formed.

Properties of covalent compounds:

- They have low melting and boiling points because weak forces hold molecules together.
- They are generally poor conductors of electricity because they have no free ions or electrons. Graphite is an exception.
- They are soluble in non-polar solvents and insoluble in water.

Allotropes of Carbon

Allotropes are different forms of the same element. The atoms are the same, but the arrangement is different. So the physical properties are different.

Allotropy

The property of an element to exist in two or more forms which have different physical properties but similar chemical properties.

Carbon has three important crystalline allotropes:

1. Diamond. Each carbon atom is bonded to four other carbon atoms in a rigid three dimensional network. There are no free electrons. Diamond is the hardest natural substance. It has a very high melting point. It is a bad conductor of electricity. It is used for cutting and drilling and in jewellery.

2. Graphite. Each carbon atom is bonded to three other carbon atoms. This forms layers of hexagonal rings. The layers can slide over each other. One electron per

carbon is free. So graphite is soft and slippery. It conducts electricity. It is used as a lubricant and as electrodes in cells.

3. Fullerene (C₆₀). It has 60 carbon atoms arranged like a football. It is also called Buckminsterfullerene. It was discovered recently and is used in advanced materials.

Note: As per the rationalised NCERT 2026-27, detailed structures of diamond and graphite are for understanding the concept. Focus on the difference in bonding and properties.

Versatile Nature of Carbon

Carbon forms more compounds than any other element. The number of carbon compounds is very large. This is due to two special features:

1. Tetravalency. Carbon has a valency of four. It can bond with four other atoms of carbon or other elements like H, O, N, S and Cl.

2. Catenation. Carbon can bond with other carbon atoms to form long chains, branched chains and rings. This self-linking property is called catenation. No other element shows catenation as strongly as carbon.

3. Ability to form multiple bonds. Carbon can form single, double and triple bonds with other carbon atoms. So it can form saturated and unsaturated compounds.

Carbon also forms very strong bonds with other elements, so its compounds are stable.

Hydrocarbons

Compounds made of carbon and hydrogen only are called hydrocarbons. They are the simplest carbon compounds.

Hydrocarbon

A compound that contains only carbon and hydrogen atoms.

Hydrocarbons are of two main types:

1. Saturated hydrocarbons. All carbon-carbon bonds are single bonds. They are called alkanes. General formula is C_nH_{2n+2} . Example: Methane CH_4 , Ethane C_2H_6 , Propane C_3H_8 . They are not very reactive.

2. Unsaturated hydrocarbons. One or more carbon-carbon bonds are double or triple bonds.

- Alkenes have at least one double bond. General formula C_nH_{2n} . Example: Ethene C_2H_4 .
- Alkynes have at least one triple bond. General formula C_nH_{2n-2} . Example: Ethyne C_2H_2 .

Unsaturated hydrocarbons are more reactive than saturated ones. They give addition reactions.

Worked Example

Identify whether C_4H_{10} and C_3H_4 are saturated or unsaturated.

C_4H_{10} fits C_nH_{2n+2} where $n = 4$: $2n + 2 = 10$. So it is an alkane and saturated.

C_3H_4 fits C_nH_{2n-2} where $n = 3$: $2n - 2 = 4$. So it can be an alkyne and is unsaturated.

Homologous Series

A homologous series is a group of compounds with the same functional group, similar chemical properties and successive members differ by CH_2 .

Homologous Series

A series of compounds in which the same functional group substitutes for hydrogen in a carbon chain, and any two successive members differ by CH_2 .

Properties of a homologous series:

- All members have the same general formula.
- Any two successive members differ by CH_2 and by 14 u in molecular mass.
- Chemical properties are similar because the functional group is the same.

- Physical properties change gradually with increase in molecular mass. Melting and boiling points increase.

Example, the alkane series: Methane CH_4 , Ethane C_2H_6 , Propane C_3H_8 , Butane C_4H_{10} , Pentane C_5H_{12} .

Worked Example

Write the next two members after C_3H_8 .

C_3H_8 is propane. Next members are obtained by adding CH_2 each time. So C_4H_{10} is butane and C_5H_{12} is pentane.

Functional Groups

A functional group is an atom or group of atoms that decides the chemical properties of an organic compound. Compounds with the same functional group show similar behaviour.

Functional Group	Formula	Example
Halo (Chloro / Bromo)	-Cl, -Br	Chloromethane CH_3Cl
Alcohol	-OH	Ethanol $\text{C}_2\text{H}_5\text{OH}$
Aldehyde	-CHO	Ethanal CH_3CHO
Ketone	-CO-	Propanone CH_3COCH_3
Carboxylic acid	-COOH	Ethanoic acid CH_3COOH
Double bond	$\text{C}=\text{C}$	Ethene C_2H_4

Nomenclature. For simple compounds, the root shows the number of carbon atoms: meth- for one, eth- for two, prop- for three and but- for four. The ending shows the bond or functional group: -ane, -ene, -yne, -ol, -al, -one or -oic acid. A position number is added where needed, as in propan-1-ol.

Worked Example

Name the compound $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$.

There are three carbons, so root is prop. Functional group is $-\text{OH}$, so suffix is $-\text{ol}$. The $-\text{OH}$ is on carbon 1. So the name is propan-1-ol or propanol.

Isomerism

Compounds with the same molecular formula but different structures are called isomers. The carbon chain can be straight or branched.

Example: Butane C_4H_{10} has two isomers. n-butane $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ is a straight chain. Iso-butane $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$ is a branched chain. Both have formula C_4H_{10} but different structures and slightly different properties.

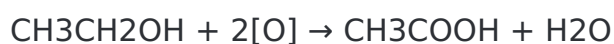
Isomerism is possible because carbon can form branched chains.

Chemical Properties of Carbon Compounds

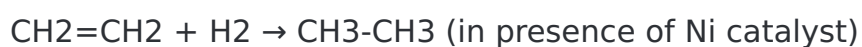
1. Combustion. Carbon compounds burn in oxygen to produce heat and light. With sufficient oxygen, hydrocarbons form carbon dioxide and water. A clean fuel may burn with a blue flame, while incomplete combustion produces a yellow, sooty flame.



2. Oxidation. Alcohols can be oxidised to acids. Ethanol is oxidised to ethanoic acid with alkaline KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$. These oxidise the alcohol by adding oxygen.



3. Addition reaction. Unsaturated hydrocarbons add hydrogen in presence of catalysts like nickel or palladium to form saturated hydrocarbons. This is used to make vanaspati ghee from vegetable oils.



4. Substitution reaction. Saturated hydrocarbons are not very reactive. But in presence of sunlight, chlorine replaces hydrogen atoms. For example, methane with chlorine in sunlight forms chloromethane.



Ethanol and Ethanoic Acid

A. Ethanol ($\text{C}_2\text{H}_5\text{OH}$).

- Ethanol is a liquid at room temperature. It is soluble in water. It is a good solvent.
- Reaction with sodium: $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$. Hydrogen gas is released.
- Reaction with acid to form ester: Ethanol reacts with ethanoic acid in presence of conc. H_2SO_4 to form ethyl ethanoate, an ester with sweet smell. This is esterification.
- Dehydration: On heating ethanol with conc. H_2SO_4 at 443 K, it gives ethene.
 $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$.
- Uses: In medicines, as a solvent, in alcoholic drinks, as a fuel additive.

Denatured alcohol is ethanol mixed with small amount of methanol or copper sulphate to make it unfit for drinking.

B. Ethanoic Acid (CH_3COOH).

- It is also called acetic acid. 5 to 8 percent solution of acetic acid in water is called vinegar.
- Pure ethanoic acid freezes in cold and looks like ice, so it is called glacial acetic acid. Its melting point is 290 K.
- It is acidic and turns blue litmus red. Reaction with base: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$.
- Reaction with sodium carbonate: $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$. CO_2 turns lime water milky.
- Esterification: $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$ (in presence of conc. H_2SO_4).
- Saponification: On heating, esters with NaOH give back alcohol and sodium salt of acid. This is used to make soap.

Worked Example

Why does ethanoic acid show acidic behaviour while ethanol does not?

Ethanoic acid can release H^+ from its $-COOH$ group in water. Ethanol has $-OH$ bonded to carbon and does not release H^+ easily. So ethanoic acid is acidic and ethanol is almost neutral.

Soaps and Detergents

Soap

Sodium or potassium salt of a long chain carboxylic acid (fatty acid). Example: Sodium stearate $C_{17}H_{35}COONa$.

How soap is made. Soap is made by heating esters of fatty acids (oils and fats) with sodium hydroxide. The reaction is called saponification. $Ester + NaOH \rightarrow Soap + Glycerol$.

Structure of soap. A soap molecule has two parts. The ionic part is hydrophilic (water loving). It dissolves in water. The hydrocarbon chain part is hydrophobic (water hating). It dissolves in oil and dirt.

Cleansing action of soap. When soap is dissolved in water, it forms clusters called micelles. The hydrophobic tails go inside the cluster and trap oil and dirt. The hydrophilic heads stay outside towards water. The micelle is then washed away with water, carrying the dirt with it.

Limitations of soap. Soap does not work well in hard water. Hard water has calcium and magnesium salts. Soap forms insoluble scum with these ions. So a lot of soap is wasted.

Detergent

Sodium salt of a long chain benzene sulphonc acid or a long chain sulphate. Detergents do not form scum with hard water.

Difference between soap and detergent:

- Soap is a salt of carboxylic acid. Detergent is a salt of sulphonic acid.
- Soap forms scum in hard water. Detergent does not.
- Soap is biodegradable. Many detergents are not fully biodegradable.
- Detergents clean better in hard water but can cause more water pollution.

Summary

- Carbon forms only covalent bonds by sharing electrons. It has valency four.
- Allotropes of carbon are diamond, graphite and fullerene. They differ in structure and properties.
- Carbon shows tetravalency, catenation and ability to form multiple bonds. So it forms many compounds.
- Hydrocarbons contain only C and H. Alkanes are saturated with general formula C_nH_{2n+2} . Alkenes (C_nH_{2n}) and alkynes (C_nH_{2n-2}) are unsaturated.
- A homologous series is a group where successive members differ by CH_2 . Members share the same functional group.
- Functional groups like $-OH$, $-CHO$, $-CO-$, $-COOH$ decide the chemical properties.
- Chemical properties include combustion, oxidation, addition and substitution reactions.
- Ethanol and ethanoic acid are important carbon compounds. Ethanol forms esters and ethene. Ethanoic acid is acidic and forms esters and CO_2 with carbonates.
- Soap is sodium salt of fatty acid. Detergent is sodium salt of sulphonic acid. Soaps form scum in hard water, detergents do not. Cleansing action is through micelle formation.

Practice Questions

1. Why does carbon form covalent bonds and not ionic bonds? Explain with electronic configuration.

2. Compare the structure and properties of diamond and graphite.

3. Draw the electron dot structures of methane, oxygen and nitrogen.
-
4. What is a homologous series? Write two members of the series of aldehydes.
-
5. Write the structural formulae of: (a) Ethanoic acid (b) Propanone (c) But-1-ene.
-
6. Explain addition and substitution reactions with one example each.
-
7. Give two chemical properties of ethanol and write the equations.
-
8. Explain the cleansing action of soap with the help of micelles. Why does soap not work in hard water?